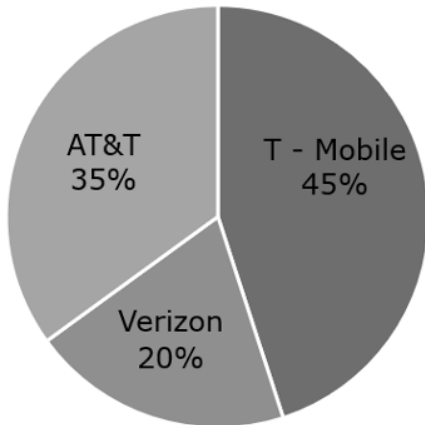


Grade 6 Accelerated
Unit 10
Lesson 10 Homework

Name Answer Key
Date _____
Period _____

550 Students in the 8th grade were polled to see what cell phone carrier was the most common in a local middle school. The middle school has a total population of 2475 students. The circle graph below shows the percentage of students that have the three major cell phone carriers: AT&T, T- Mobile and Verizon.
Use the circle graph to answer questions 1-3.

Student Cell Phone Carriers



1. To predict how many students with Verizon as a carrier, the student council president wants to use the scale factor method. What scale factor should she use to make this prediction?

$$\frac{2,475}{550} = 4.5$$

2. Explain whether the equivalent factor or percent method be used to predict the number of students with AT&T as a carrier should the equivalent factor or percent method be used?

Answers may vary. Percent method because they're given in the circle graph.

3. Predict the number of students that have T-Mobile as a cell phone carrier.

$$0.45 (2,475) = 1,113.75 \approx 1,114 \text{ students have T-Mobile.}$$

Tyrone wants to know about the color distribution of Skittles in his candy jar. Without looking he picked out one Skittle at a time, recorded it, and put them back into the jar. Tyrone picked 10 green, 6 purple, 12 red, 8 yellow, 4 orange after 40 picks. A total of 220 Skittles are in the jar. Use Tyrone's results to answer questions 4-5.

4. What fraction of Skittles are green, purple, red, yellow and orange?

$$\text{Green: } \frac{10}{40} = \frac{1}{4}$$

$$\text{Red: } \frac{12}{40} = \frac{3}{10}$$

$$\text{Orange: } \frac{4}{40} = \frac{1}{10}$$

$$\text{Purple: } \frac{6}{40} = \frac{3}{20}$$

$$\text{Yellow: } \frac{8}{40} = \frac{1}{5}$$

5. Assuming his sample is a true representation of Skittle distribution, how can Tyrone predict the estimated number of Skittles in the jar that are red?

Answers may vary. Multiply the fraction of red Skittles by the total number of skittles in the jar.

The cheerleading team at a local school cannot attend all of the games for all sports teams in the month of February due to conflicting schedules. The Athletic Director, Mrs. James, wants to prioritize the games the team will cheer at to boost morale of the teams. Mrs. James asks 150 students to pick the sport the cheerleading team should cheer. The results were recorded in the table below. The school has a total of 1150 students. Use the results of the sample to answer questions 6-10.

Sport	Number of Times Chosen	Ratio of Students that Picked Each Sport
Baseball	25	$\frac{25}{150} = \frac{1}{6}$
Basketball	55	$\frac{55}{150} = \frac{11}{30}$
Volleyball	40	$\frac{40}{150} = \frac{4}{15}$
Soccer	30	$\frac{30}{150} = \frac{1}{5}$

6. Find the ratio of students in the sample that picked each sport and complete the table.

7. Predict the number of students in the school that want the cheerleading team to perform at the baseball games (based on sample results) using the percent method.

$$\frac{1}{6} (1,150) = 191.\bar{6} \approx 192 \text{ students}$$

8. Explain which is the best method to predict the number of students in the school that want the cheerleading team to perform at the soccer games (based on sample results).

Answers may vary. Percent strategy because we have the fraction form of the percent.

9. Mrs. James uses the scale factor method to predict the number of students in the school that want the cheerleading team to cheer at basketball games. She uses a scale factor of 0.13. Did Mrs. James use the right scale factor? Explain.

$$\text{No because } \frac{\text{population}}{\text{sample}} = \frac{1150}{150} = 7.\bar{6}$$

10. The cheerleading team prefers to cheer at indoor sports. Predict the amount of students that prefer the cheerleading team to perform at the indoor sports.

$$\frac{55+40}{150} = \frac{95}{150} (1,150) = 728.\bar{3} \approx 728 \text{ students}$$